



■ CS177 Bioinformatics: Software Development and Applications

Lecture 17: Metabolomics

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Spring, 2024

Slides adapted from talk of Jack Gardiner “**Metabolomics - A Very Brief Introduction Jack Gardiner**” (March, 2021) <https://www.youtube.com/watch?v=DFR3Syq1lpU>

Learning objectives



- At the end of this lecture, you should be able to:
 - Define what is metabolomics
 - Describe the workflows of metabolomics studies
 - Describe three example metabolomics databases





Overview of Metabolomics



Metabolomics defined



- **Metabolomics** is the “systematic study of unique chemical fingerprints that specific cellular processes leave behind” ¹
- Generally speaking, metabolomics is limited to molecules 1.5 kDa (1500 g/mol) or less
 - Metabolomics is sometimes referred to as “small molecule profiling”
- Remember that a metabolic profile represents but a single snapshot in time. The metabolome is a dynamic entity and change from second to second
 - More recent efforts focus on **Real Time Metabolomics** which can measure hundreds of metabolites in a living cell in real time

¹ Davis, B (April 2005). "Growing pains for metabolomics". *The Scientist*. **19** (8): 25–28.



What molecules are measured in Metabolomics?



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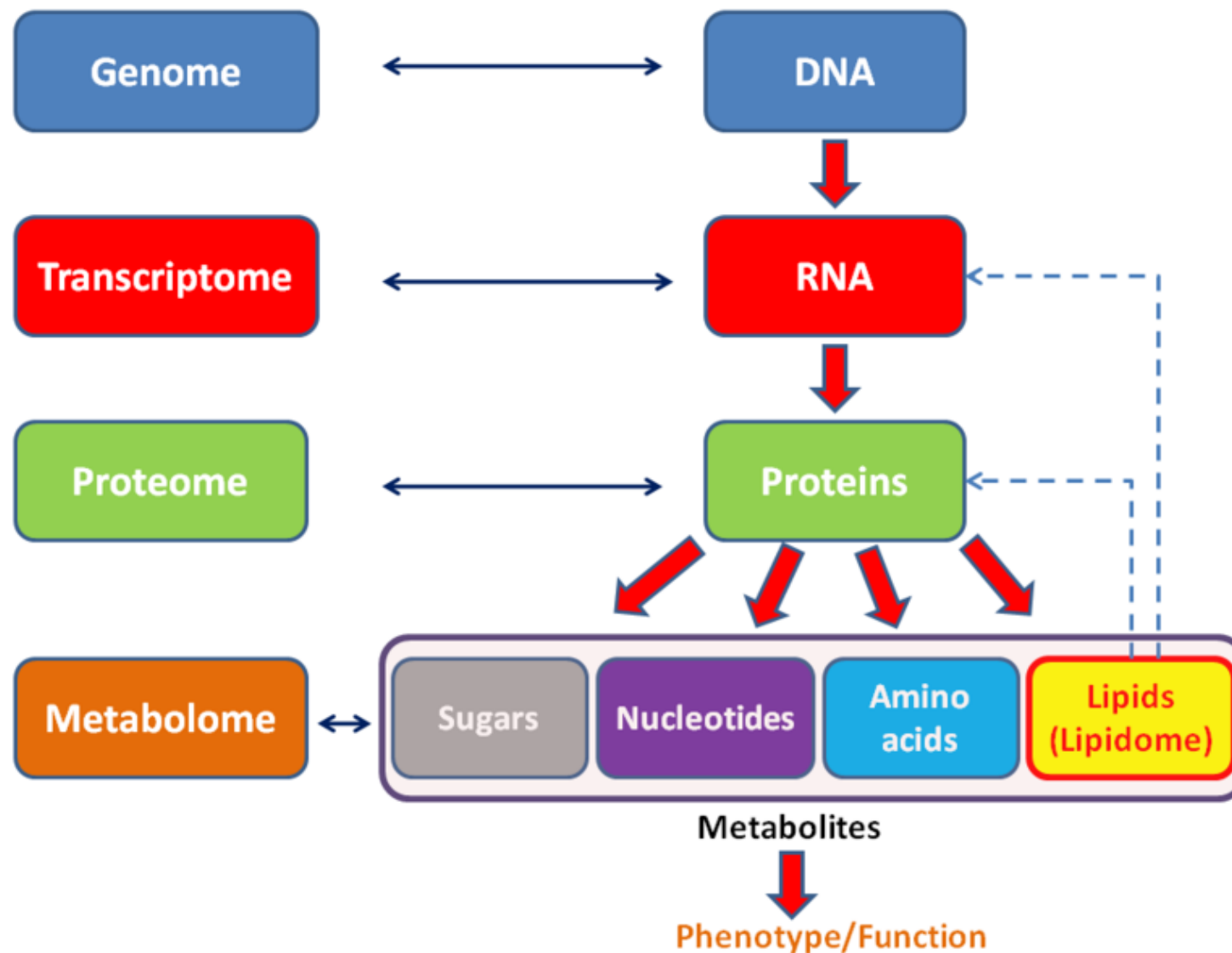
- **Endogenous** (produced **within** the organism): amino acids, lipids, cofactors, nucleotides, sugars, hormones, etc.
- **Exogenous** (produced **external** to the organism): drugs, toxins, environmental contaminants, pesticides, herbicides, etc.



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A Geneticist's view of Metabolomics



<https://en.wikipedia.org/wiki/User:Lmaps>





Relationships among the omics



- *Genome can tell what could happen,*
- *Transcriptome can tell what appears to be happening,*
- *Proteome can tell what makes it happen, and*
- *Metabolome can tell what has happened and what is happening*

Dettmer K. et al (2007). "Mass spectrometry-based metabolomics". Mass Spectrometry Reviews. **26**(1): 51–78.



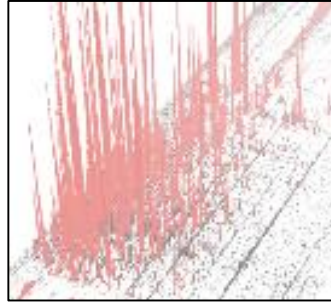


Metabolomics: Two basic approaches



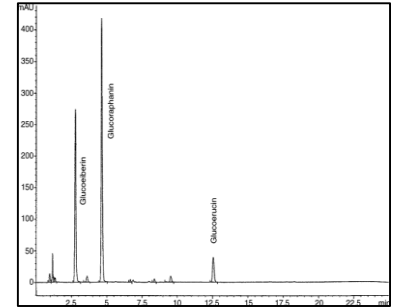
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Untargeted: Shotgun Profiling



- Provides the opportunity to see everything
 - Many known, and even more unknowns
 - Relative abundances of hundreds of compounds
 - Data analysis is more difficult
- Potentially a powerful phenotyping tool
 - Need to have differences between bio samples for GWAS/QTL mapping

Targeted: Classical Analytical Biochemistry



- Can optimize conditions for specific metabolites
 - Flavonoids, carotenoids, hormones, etc.
 - Can be quantified
 - Data analysis is easier
- Downside: standards are needed
 - The number of known compounds is limited





Bio samples vary greatly as to how many metabolites they contain



Bio Sample	Total Number of Metabolites Detected	Percent Quantified
Sweat	91	86%
Cerebrospinal Fluid	445	99.5%
Saliva	1245	71%
Urine	2603	68%
Blood	3678	84%
Feces	6810	26%

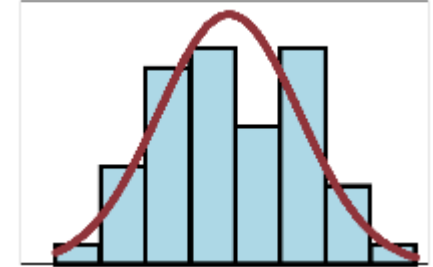




Statistical and sampling considerations



- Experimental design is critical
 - 3 repeats minimum, 5 repeats preferred
 - Sample pooling is beneficial
- Biological are preferred over technical repeats
 - In general, biological variance exceeds technical variance
- Many free software packages are available
 - PCA is often used in untargeted metabolic profiling
 - Machine learning methods can improve metabolite identification
- Substantial metabolite variation is normal even in carefully collected samples
 - Not all metabolites are tightly regulated. This can hamper statistical analysis

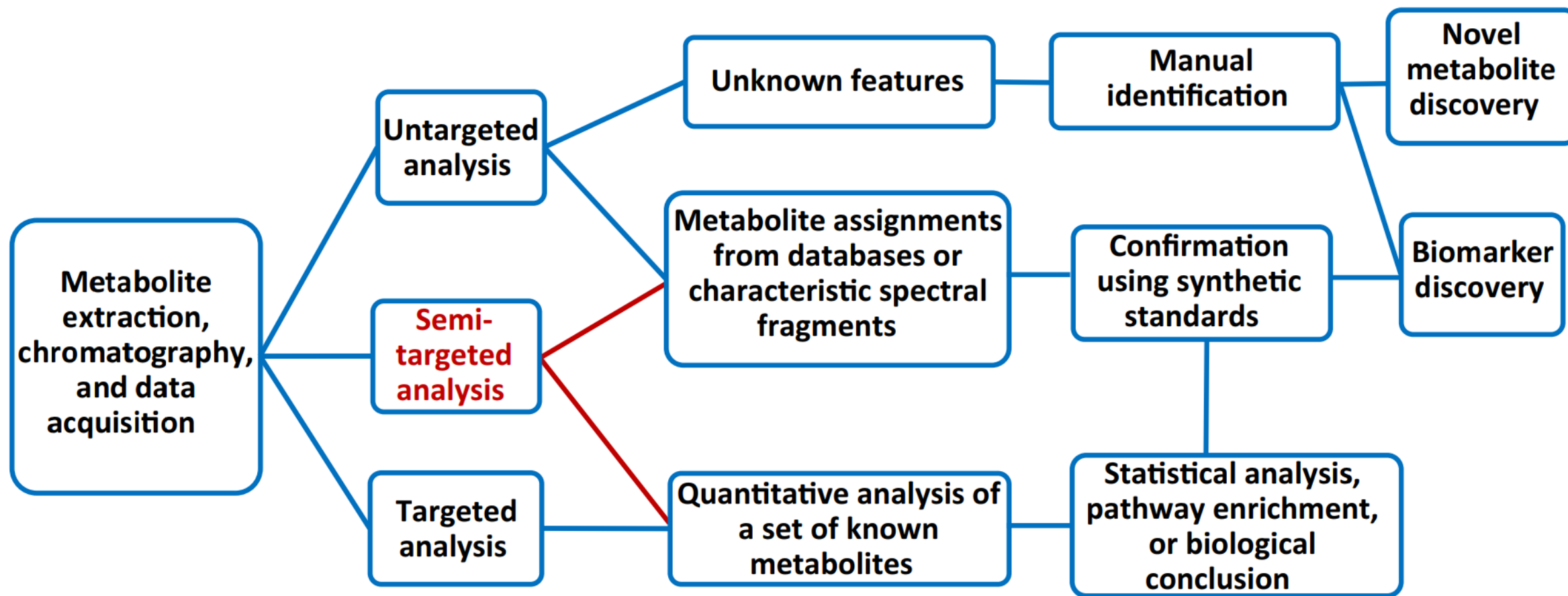




Metabolomics Workflows



General schema of metabolomics workflows

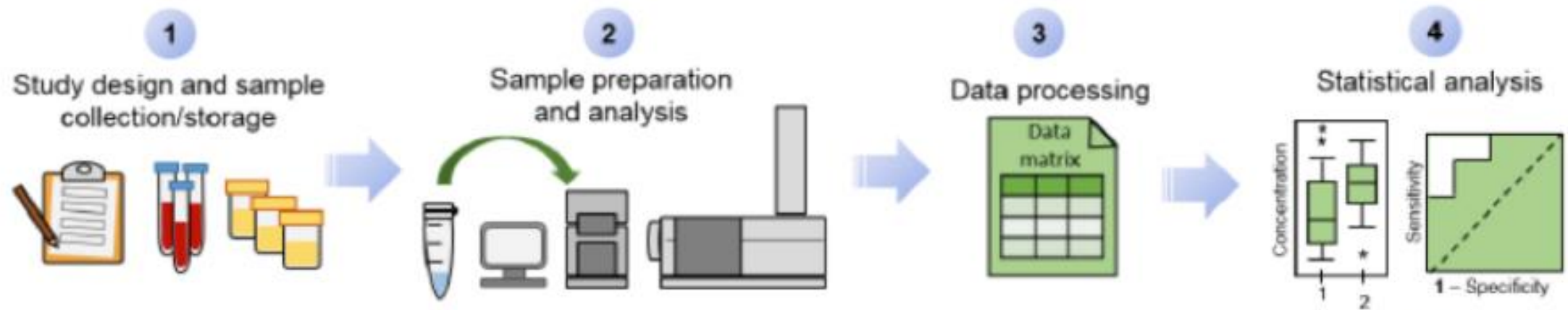


Xiaojing Liu and Jason W. Locasale (2017). "Metabolomics: A Primer". Trends in Biochemical Sciences, Vol. 42, No. 4, pp. 274-284.

Metabolomics workflow



- Sample Extraction/Collection
- Separation
- Detection
- Data Extraction



■ Sample Extraction (SE)



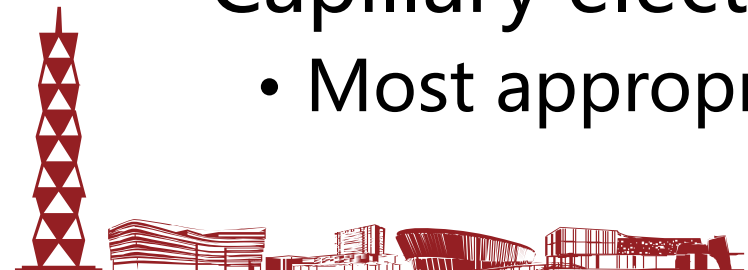
- An ideal SE should be:
 - selective
 - simple and fast
 - reproducible
 - metabolite quenching to capture metabolites in native state
- Samples are generally extracted with an organic solvent and concentrated
- Internal standards are often added at the SE preparation stage
 - Aids downstream analysis especially for targeted approaches
- A “**derivatization**” step is often included
 - Improves chromatography or prevents degradation
 - Adds additional steps





■ Sample separation

- Sample separation into individual components is usually achieved by one of three methods
- Gas chromatography (GC)
 - Ideal for small and volatile molecules
 - Greater sensitivity requires additional step of derivitization
- High performance liquid chromatography (LC/HPLC)
 - Most common method, no derivitization required
 - Wider range of molecules can be analyzed with greater sensitivity
- Capillary electrophoresis (CE)
 - Most appropriate for charged molecules





Sample detection



- After separation, individual metabolites are *usually* detected by Mass Spectrometry (MS) which measures its mass to charge ratio (m/z). NMR is sometimes used but it lacks the sensitivity of MS
- Many types of MS (ESI-MS, MALDI-TOF-MS)
- Common to all MS technologies are
 - an ionization source,
 - mass analyzer, and
 - a detector
- Many variations on each of these three components



Sample detection (continued)

- Hard vs. soft ionization approaches
 - Soft: molecules are left intact
 - Hard: molecules are fragmented to produce a more detailed fingerprint
- Tandem MS uses multiple rounds of MS
 - High resolution fingerprints are produced and compared to standards
 - Useful in the identification of unknown metabolites
 - **METLIN Metabolite and Chemical Entity Database** contains over 850,000 known standards used to assist the identification of known and unknowns

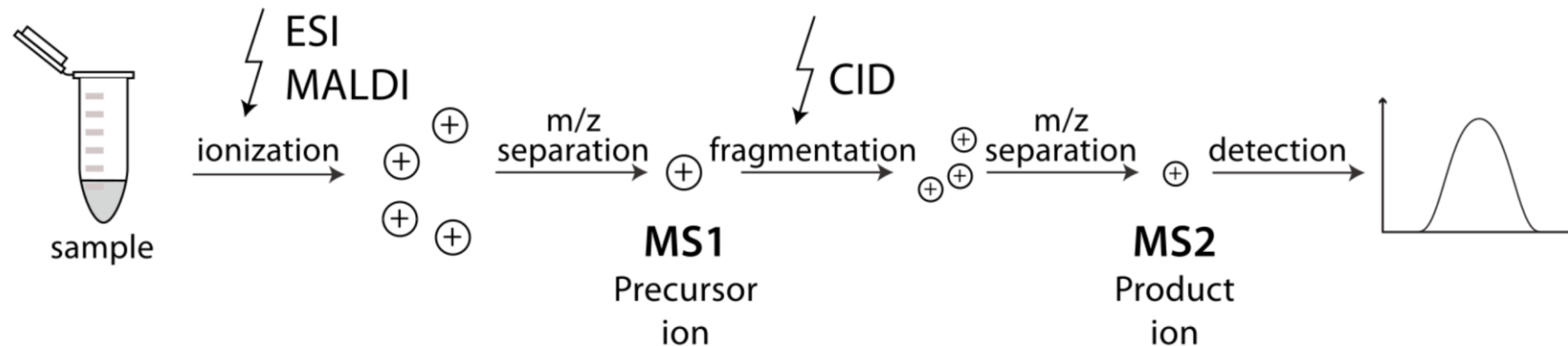


Figure source: <https://en.wikipedia.org/wiki/File:TandemMS.svg>



Data extraction and analysis



- Data under various experimental conditions are usually extracted in the form of mass spectra values (m/z) or metabolite features
 - Lots of free software available for identifying metabolites on the basis of m/z or retention time values
- Once the data have been assembled into a treatment vs. compound table, various statistical or data reduction methods can be applied to ferret out statistically significant, meaningful biological differences.

You do not need to know the identity of each compound at this point

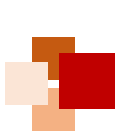
- Principal Component Analysis (PCA) is often used when the identities of molecules of interest are unknown (untargeted)
- Linear models can be used but can be impacted by multicollinearity
- Univariate and multivariate statistics are often used but care must be taken to validate models and correct for multiple comparisons
- Machine Learning approaches are increasingly used to apply retention time to the prediction and identification of metabolites in a complex mixture



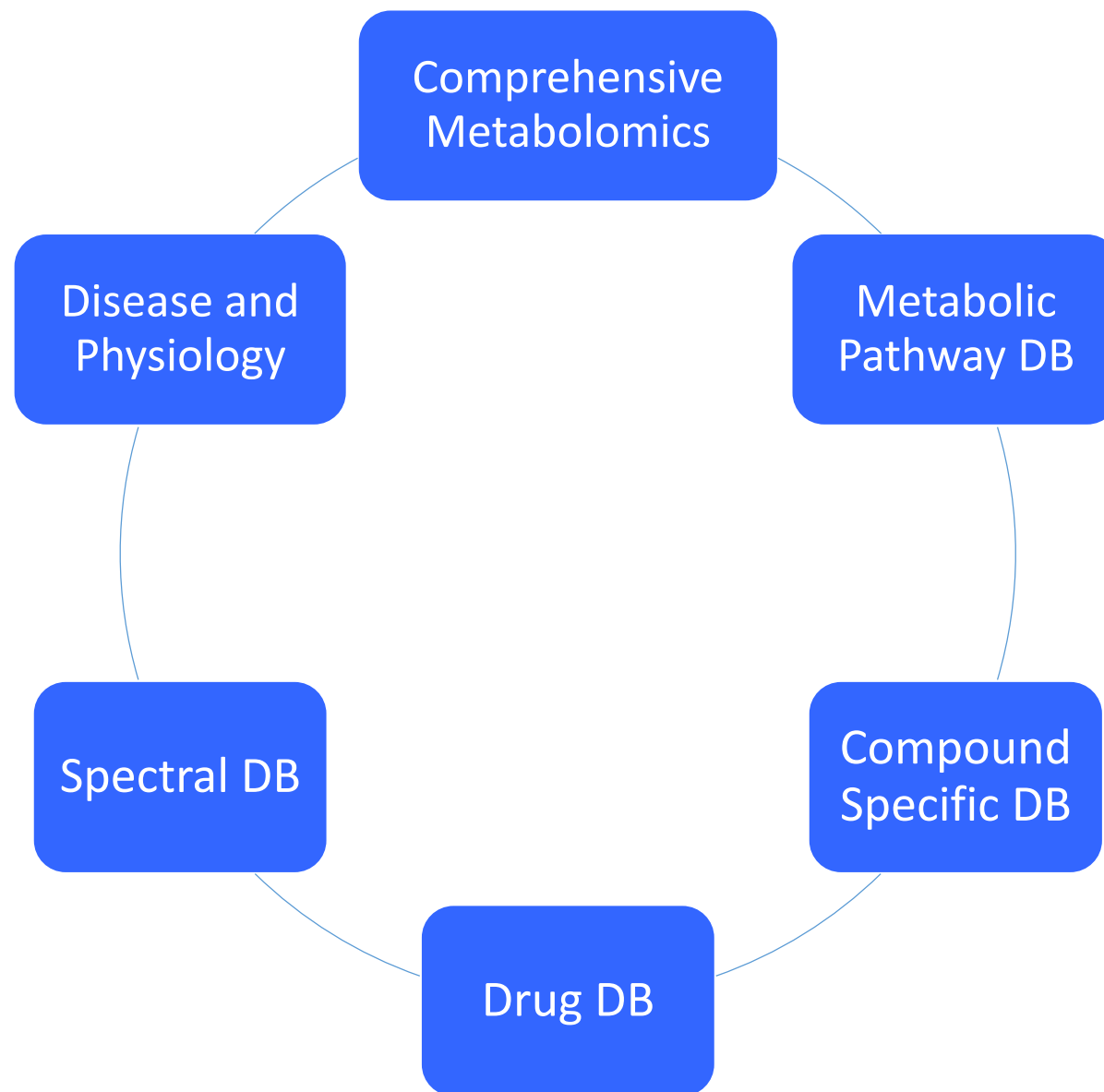


Metabolomics Databases





Metabolomics/Metabolic Databases



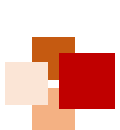


Three examples of very different metabolic databases



- Human Metabolome Database (HMDB)
 - Human centric
- MetaboLights
 - Making data available
- METLIN
 - Metabolite identification





Human Metabolome Database (HMDB)



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- HMDB (www.hmdb.ca/) contains freely available verified information on ~114,000 metabolites found in the human body
 - Each “Metabocard” entry contains 130 data fields with 2/3 being dedicated to Clinical/Chemical data and 1/3 to enzymatic and biochemical data
- HMDB is part of a suite of four additional databases:
 - DrugBank (www.drugbank.ca) information on ~2280 drugs and drug metabolites
 - T3DB (www.t3db.org/) information on 3670 toxins and environmental pollutants
 - SMPDB (<http://www.smpdb.ca>) information on pathway diagrams for 25,000 metabolic and disease pathways
 - FooDB (www.foodb.ca) information on 28,000 foods and food additives
- Both simple and sophisticated searches
 - Text search facilitates sophisticated searches of HMDB text
 - Sequence search allows BLAST searches of over 5700 DNA and protein sequences
 - Advanced search allows structured searches of various combinations of subfields
 - MS search allows searches of LC/MS spectra and 1D and 2D NMR spectra



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MetaboLights



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- **MetaboLights** (<https://www.ebi.ac.uk/metabolights>) is an all encompassing database focused on metabolomics experiments and the data derived from them. It is both a repository and a resource for curated metabolomics data
 - Cross-species and cross technique
 - Includes metabolite structures, reference spectra, cellular locations, roles, and experimental metadata
- MetaboLights allows users to:
 - Submit and browse metabolomics data (studies)
 - 778 studies submitted to date (before March, 2021)
 - Can search by species, metabolite, or technique
 - Receive stable identifiers for publication references
 - Share data with peer reviewers or collaborators
 - Download complete studies as zipped archive
- MetaboLights provides online training and education resources in metabolomics
 - Both basic and advanced training are offered online

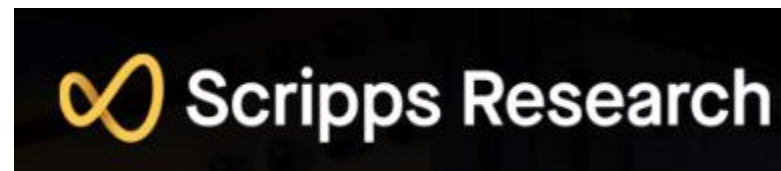


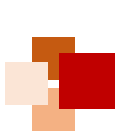


METLIN Metabolite and Chemical Entity Database

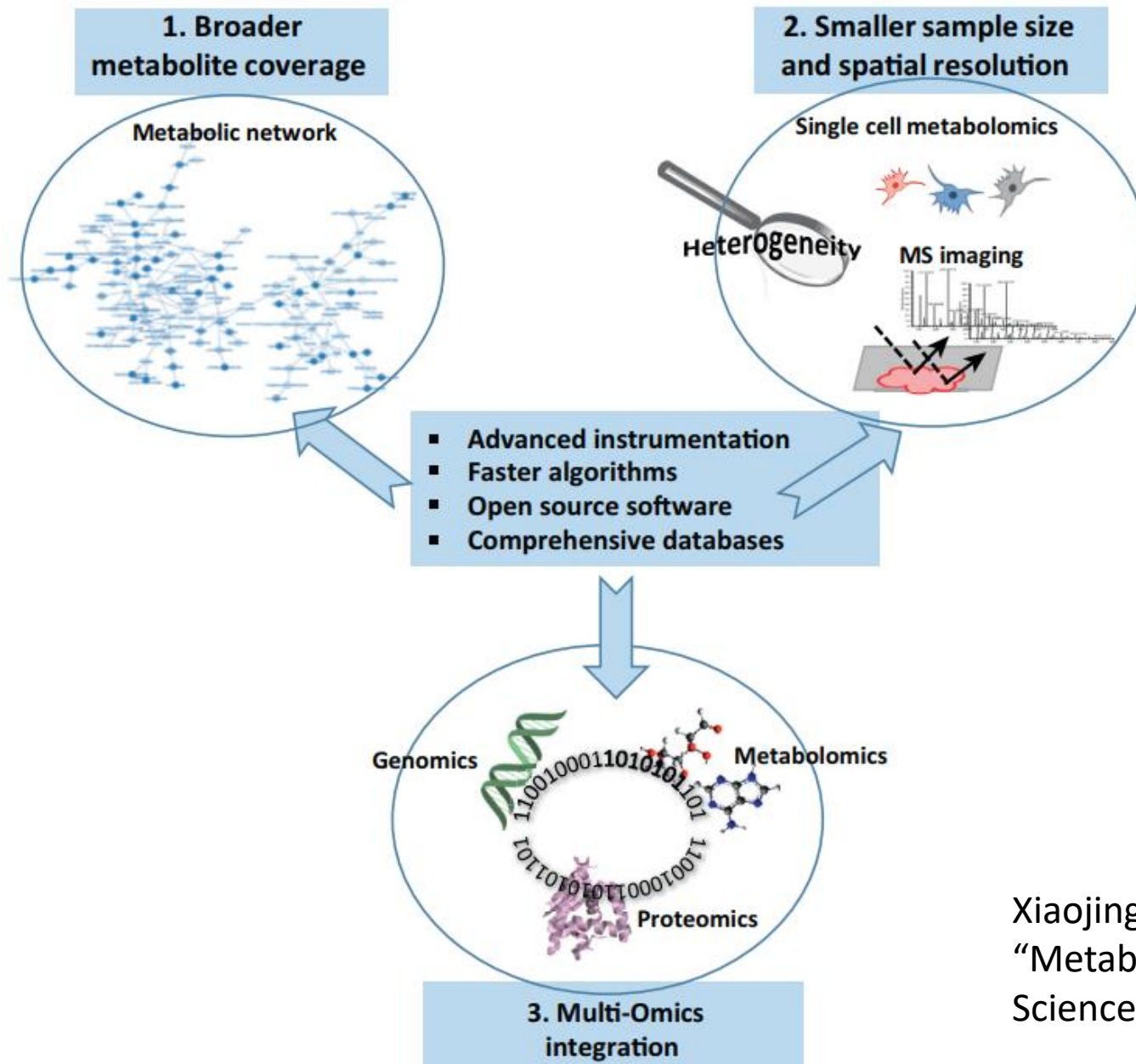


- METLIN (<https://metlin.scripps.edu>) is the largest repository devoted to **tandem mass spectrometry**. It is a broad chemical and biological resource that contains data on over 850,000 standards
 - All standards are characterized at multiple collision energies in both positive and negative ionization modes. This results in a very detailed fingerprint
 - METLIN can be searched based on tandem MS spectra, mass, chemical formula, or structure
 - Very powerful at the identification of known and unknown metabolites
 - Cloud-based and free to public and private researchers worldwide
- Developed and maintained by the Siuzdak Laboratory (<https://masspec.scripps.edu/>) at Scripps Research





Future trends in Metabolomics



Xiaojing Liu and Jason W. Locasale (2017).
“Metabolomics: A Primer”. Trends in Biochemical
Sciences, Vol. 42, No. 4, pp. 274-284.



Summary



- Metabolic profiling technologies continue to evolve
 - MS Detection is an active and evolving discipline
 - Metabolite identification remains challenging
 - Some estimates are that only 1% of molecules are identified
- Experimental design is, and will remain critical
 - Sample pooling and replication are critical
 - Collect as much metadata as possible
 - Consult a statistician **BEFORE** the experiment planning stages
- Phenotyping and metabolic profiling are a powerful combination
 - Variation is needed. Diversity panels (GWAS) are preferred over bi-parental populations (QTLs)
 - Untargeted approaches are challenging but will yield the most information
- Databases play a key role now and in the future
 - Room for everyone





References



- Xiaojing Liu and Jason W. Locasale (2017). “Metabolomics: A Primer” . Trends in Biochemical Sciences, Vol. 42, No. 4, pp. 274-284.
- Cambiaghi A, Ferrario M, Masseroli M (2017). Analysis of metabolomic data: tools, current strategies and future challenges for omics data integration. Briefings in Bioinformatics, 18(3):498-510.

